

Project Report

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1 Introduction

In this project, our objective was to model a healthcare scenario using an ontology-driven conceptual modeling approach. The targeted domain was the process surrounding General Practitioner (GP) offices in the Netherlands. This included the appointment-making process, the organization of the healthcare system, focusing on the GP office, and the role of pharmacists. By analyzing this domain and generating OntoUML models, we seek to understand the concepts, relationships, and constraints of this domain. For a better understanding of the domain, three diagrams were made to explain the process thoroughly: calling for an appointment, organization at the GP's office, and the pharmacy process.

2 Domain

2.1 Domain Description

The domain we would like to represent as a UML diagram is the process of making the appointment, going to the scheduled appointment, and further medical examination practices at the General Practitioner's office in the Netherlands. The domain description was inspired by the following online guidelines, [2] and [1].

2.2 Calling for Appointment

2.2.1 Description

The following model 1 depicts the situation when a registered patient calls the GP office to schedule an appointment.

Registered patient (Registered Patient) is a person that has registered in the GP office (MedicalPractice(GPOffice)) and can have a registration (Registration) at only one office. On the other hand, a GP office can have many

registered patients. Registered patient can also be an adult (Adult) or a minor (Minor), where minor is defined by the law in the Netherlands, as a person that is younger than 16 years which is crucial for the situation when they are seeking medical attention. In addition, the registered patient can be ill (Ill Person) or healthy (Healthy Person).

When scheduling an appointment, the registered patient will call (Call) the assistant at the GP office (GPAssistant). The registered patient, had maybe never called the assistant, but assistant has definitely received at least one call at their workplace. In case the patient is a minor (Minor Registered Patient), a legal guardian (Legal Guardian) who is an adult, has to provide Parental Consent which can be given multiple times, depending if the minor had to seek medical attention on several occasions.

Registered patient is given an appointment (Appointment) with a professional who is suitable to their case (Healthcare Professional). A healthcare professional has to work for at least one GP office, but can also work for more. The GP assistant is considered a healthcare professional, and therefore is bound to the GP office through an Employment Contract.

2.2.2 Diagram

See *figure 1* for the diagram.

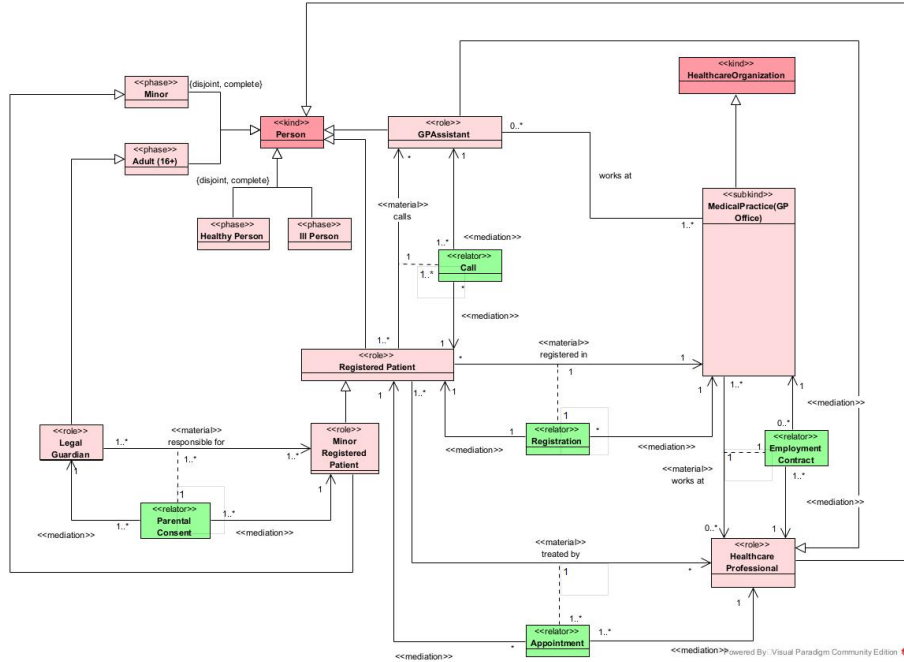


Figure 1: Registered patient calling the GP Office to schedule an appointment

2.2.3 Patterns

In the diagram showing a registered patient calling the GP office, the following patterns were observed: 2 phase patterns (*figure 2*), 5 relator patterns (*figure 3*) and 4 role patterns (*figure 4*).

The two **Phase** patterns depict the phases of a Person, Minor and Adult (16+), and Healthy Person and Ill Person. For both patterns it is made clear that they are disjoint and complete.

The five **Relator** patterns include all of the relators in the model, namely Parental Consent (Legal Guardian and Minor Registered Patient), Call (Registered Patient and GPAssistant), Registration (Registered Patient and GPOffice), Employment Contract (GPOffice and Healthcare Professional), and Appointment (Healthcare Professional and Registered Patient).

Finally, the four **Role** patterns include the role of GPAssistant and its association to GPOffice, then in a similar way for Healthcare Professional. The role of Legal Guardian is included in a pattern together with Minor Registered Patient through their association, and so is a Registered Patient with the GPAssistant.

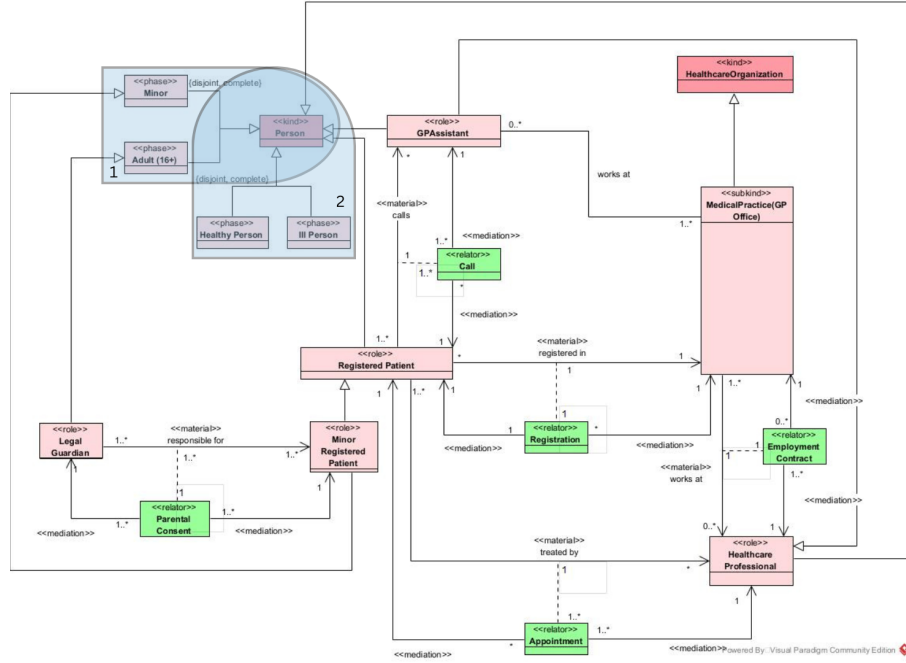


Figure 2: Phase patterns in the diagram where a registered patient calls the GP office

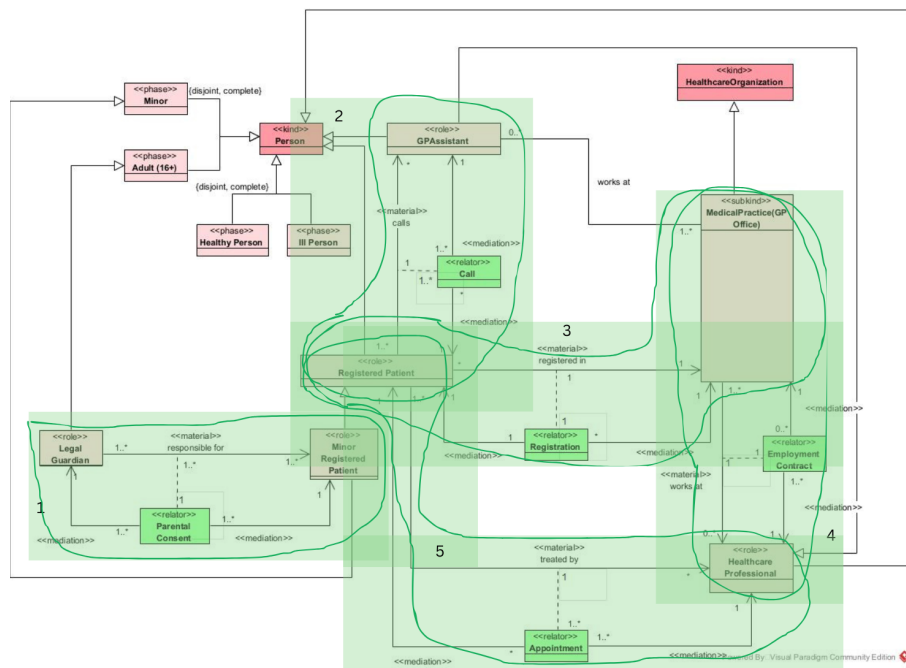


Figure 3: Relator patterns in the diagram where a registered patient calls the GP office

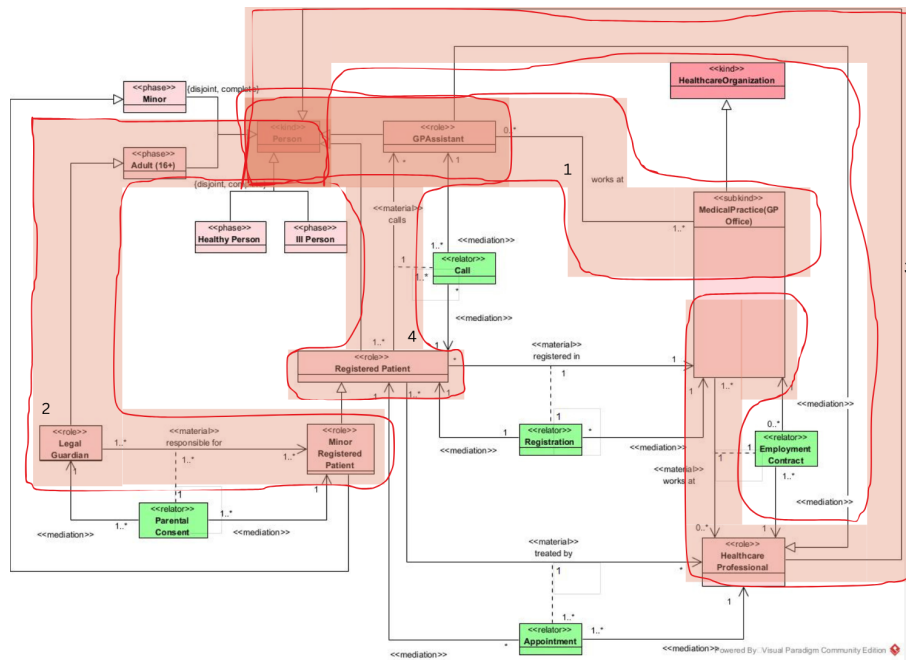


Figure 4: Role patterns in the diagram where a registered patient calls the GP office

2.3 Organization at the GP's office

2.3.1 Description

This UML model (*Figure 5*) is based on different types of Healthcare organizations, such as healthcare centers (hospitals), medical practices (GP offices), and GPPost (a central office for emergencies). A GP practice can be run by a single GP (SoloPractice), or by at least two (GroupPractice), in which case they share facilities, staff and patients. At the GP office, there are various healthcare professionals with different specializations. Each professional has a contract with their workspace and provides an availability used for GPPost.

The first contact at the GP office is the GP Assistant (Dokterassistent). They have the role of taking care of the administration, arranging the consultation hours, and providing the patients with information. They handle a variety of minor treatments and tests such as pap smears, blood pressure checks, injections, etc.

If patients need a more careful examination, they are sent to a practice assistant somatic (POH-S) or a mental health practice assistant (POH-GGZ). These roles handle many tasks, including medical and nursing work. A POH-S offers somatic care for cases as diabetes, asthma, and lung diseases, while a POH-GGZ specializes in mental healthcare and treats psychological issues. The latter type of POH is categorized as other professionals, as they have a special temporary contract with the practice and can work in multiple locations.

Most General Practitioners are part of a network of practices that provide after-hours (evening, night, or weekend) care. GPs are usually rotated for this position and therefore have a temporary employment contract based on their work shifts.

2.3.2 Diagram

See *Figure 5*.

2.3.3 Patterns

In *Figure 6*, there are two subkind-generalization sets collecting disjoint and complete sets of subkinds that specialize the same universal (HealthcareCenter, GPPost, MedicalPractice specializing HealthcareOrganization; GroupPractice, SoloPractice specializing MedicalPractice).

Role Pattern are types whose identity and characteristics depend on the relational context. In *Figure 7*, there are a few such patterns: HealthcareProfessional-GP-EmploymentContract-MedicalPractice, HealthcareProfessional-GPAssistant-EmploymentContract-MedicalPractice, HealthcareProfessional-POH-S-EmploymentContract-MedicalPractice, HealthcareProfessional-PracticeNurse-EmploymentContract-MedicalPractice, HealthcareProfessional-OtherProfessional-CollaborationAgreement-MedicalPractice, OtherProfessional-POH-GGZ, CollaborationAgreement-MedicalPractice, GP-OnCallGP-TemporaryEmploymentContract-GPPost.

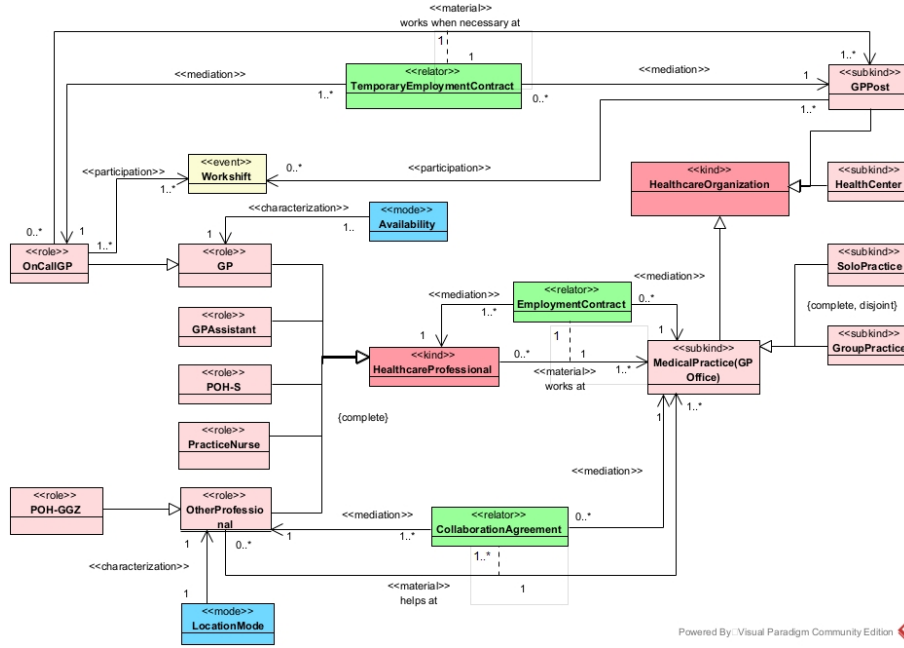


Figure 5: Staff Organization at the Dutch GP's Office

In the case of the **Relator Pattern** (Figure 8), there are three binary material relations with the two universal relata involved (OtherProfessional-MedicalPractice, HealthcareProfessional-MedicalPractice, OnCallGP-GPPost). As this figure illustrates, the cardinality constraints of the material and derivation relation are restricted by the cardinality constraints of the mediation relations in the pattern.

2.4 Pharmaceutical Process

2.4.1 Description

The pharmaceutical UML model is based on the process of getting medication, starting at the prescription from a general practitioner. The foundational types in this model are of the kinds Person, Organization, Device and Medication. The different processes within the model are described in the following sections.

General Practitioner and Prescription The process starts when a General Practitioner (role), creates a Prescription. The Prescription itself is modeled as a relator, and mediates a General Practitioner, the Patient (Person), and a Medication Collection (historicalRole).

Pharmacy The Pharmacy is modelled as a subkind of Organization, and employs individuals via the relator Employment. Employed people can play the role of Pharmacy Employee, which is further specialized into Pharmacist and

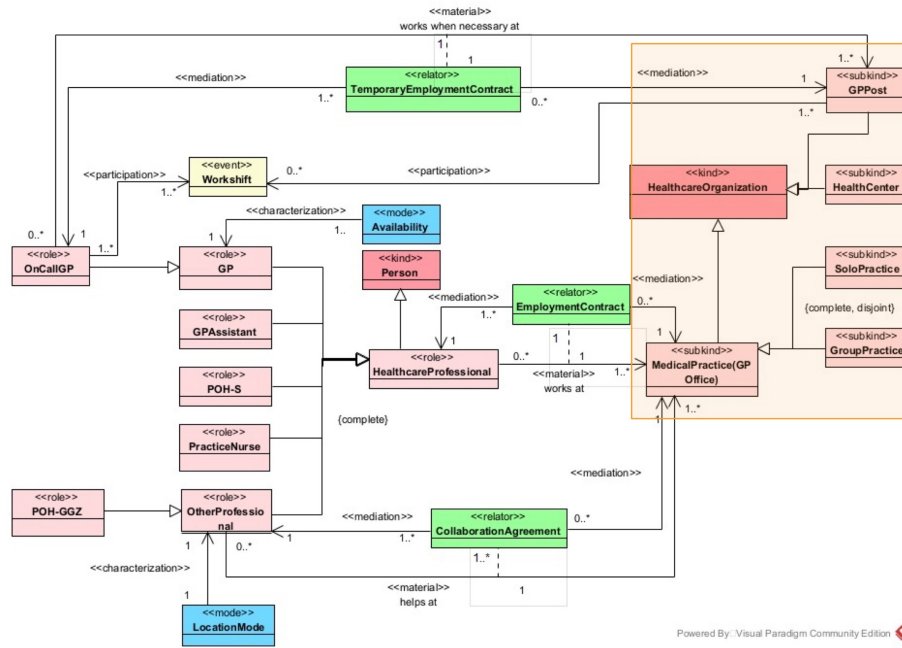


Figure 6: Subkind Pattern - Staff Organization at the Dutch GP's Office

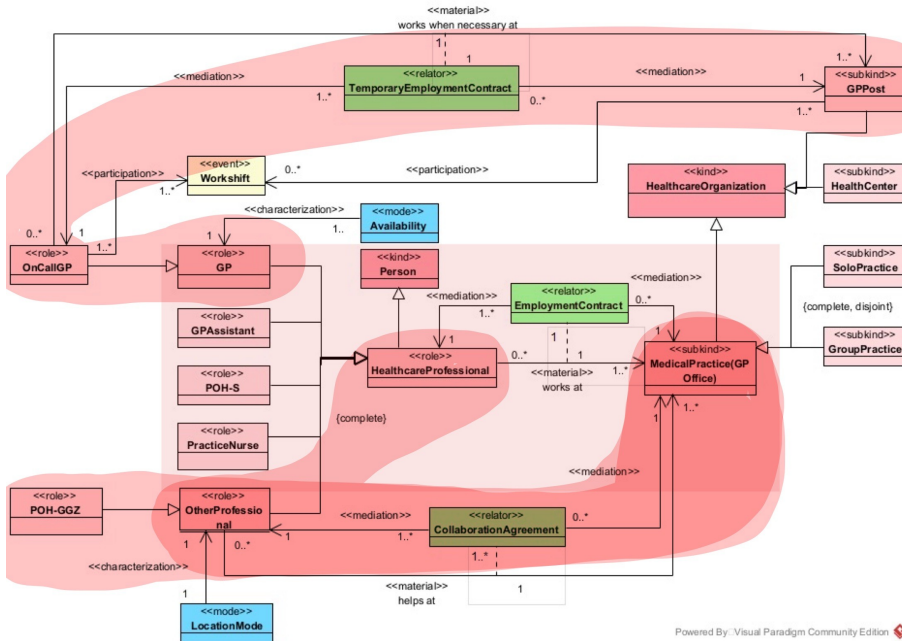


Figure 7: Role Pattern - Staff Organization at the Dutch GP's Office

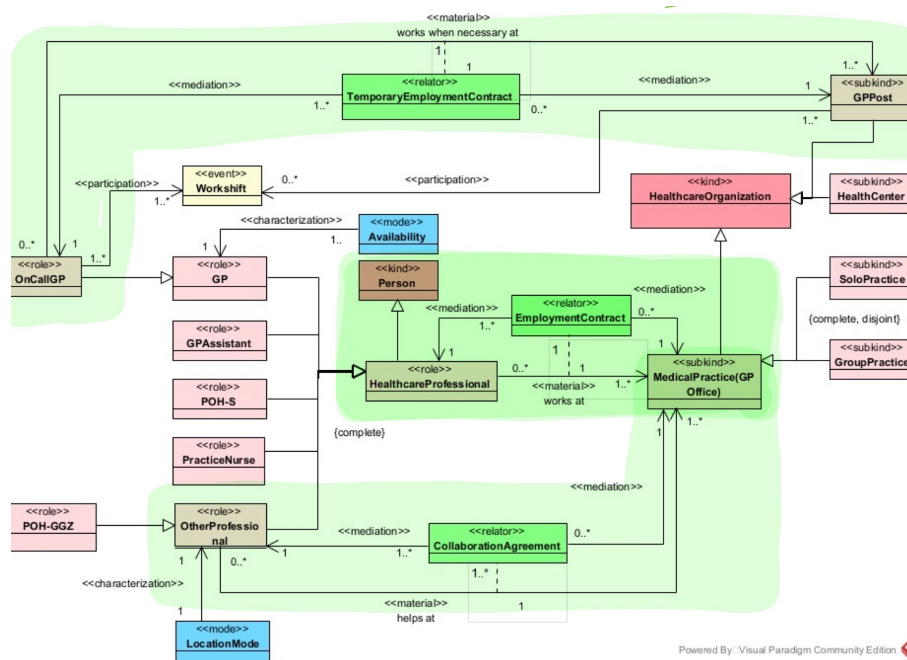


Figure 8: Relator Pattern - Staff Organization at the Dutch GP's Office

Pharmacy Assistant. These are historical roles, in order to keep track of the exact pharmacist that approved a prescription and the employee that dispensed medication.

Prescription Validation Before a collection of medication can be dispensed, a Prescription Validation occurs. This relator mediates between a Historical Pharmacist and the corresponding Prescription. This validates that the medication was correctly prescribed by the GP and appropriate for dispensation. This validation steps links a person to the accountability of approving the prescription.

Dispensation The Dispensation of medication is modelled as an event. It has two specializations, namely In Person Dispensation and Machine Dispensation. These are disjoint and complete subtypes of Dispensation. They present the actual delivery of medication to the patient. This can be either in person with a Historical Pharmacy Employee, or via a Collecting Machine (subkind of kind Device). The Medication Collection and the Patient participate in this event.

Medication Medication is modeled as a subkind with a link to its kind Medication Type. Each realization of medication is a Medication Instance, modelled as a historicalRole. Each instance has a dosage and amount attribute. These instances are grouped in a historical role called Medication Collection, which is of kind Medication Box.

Medication Inventory To capture the inventories of pharmacies, a relator called Pharmacy Stock is used. It mediates between a Pharmacy and a Medication, and records the attributes stockQuantity and restockDate. This ensures that each pharmacy can maintain their own pharmacy stocks for each medication type it has.

2.4.2 Diagram

See Figure 9.

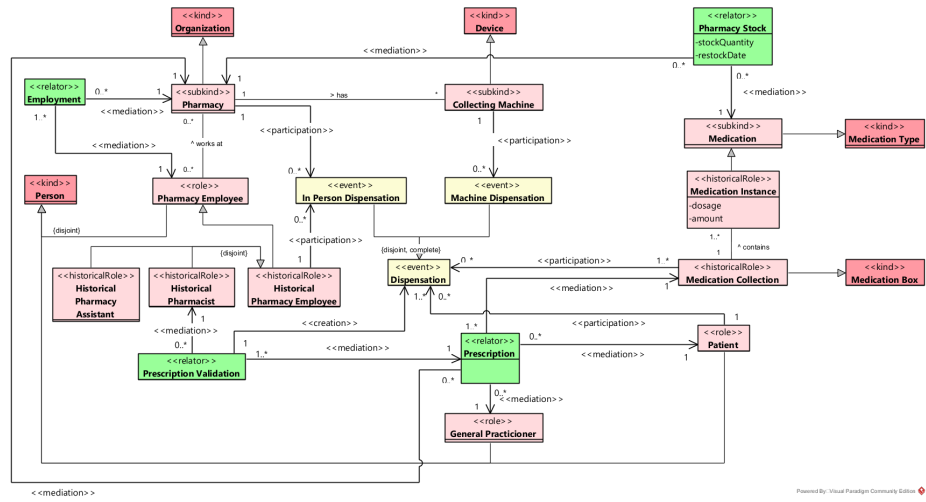


Figure 9: Pharmacy Model

2.4.3 Patterns

Several patterns can be identified in this model. This section contains an overview of all patterns in the pharmacy model.

Subkind Pattern (Figure 10) The model contains three subkind patterns. The subkind Pharmacy, is of kind Organization. The subkind Collecting Machine, is of kind Device. And the subkind Mediation, is of kind Medication Type.

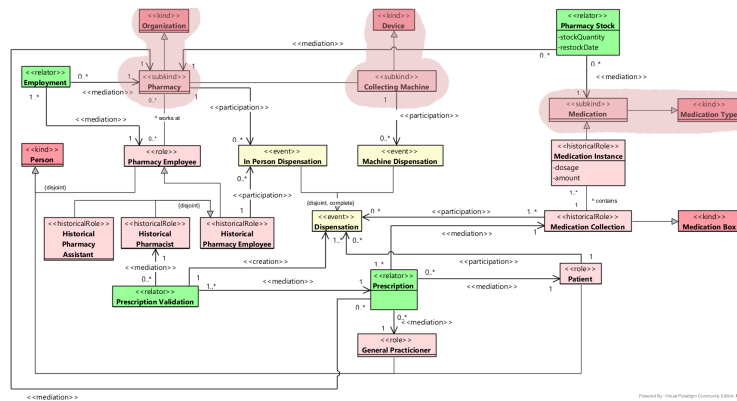


Figure 10: Pharmacy Model - Subkind Pattern

Role Pattern (Figure 11) Several role patterns are present in the model. Many roles are connected to the kind Person: Pharmacy Employee, Historical Pharmacy Employee, Historical Pharmacist, Historical Pharmacy Assistant,

General Practitioner, and Patient. Next to people, there are also roles for Medication Instance and Medication Collection.

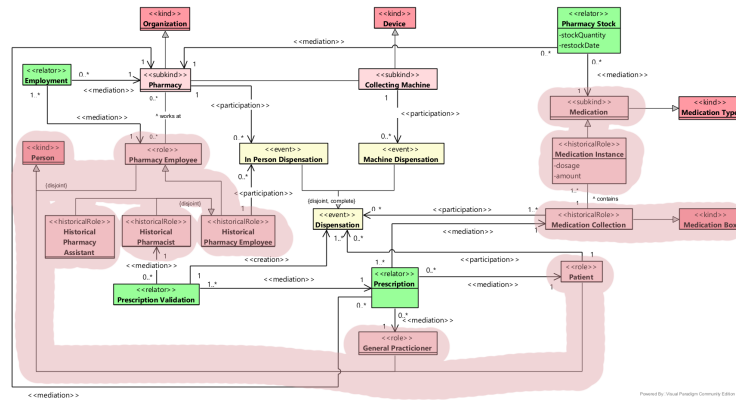


Figure 11: Pharmacy Model - Role Patterns

Role Pattern (Figure 12) The relator pattern is often used in this model. A summation:

- Employment: to model the employment of a pharmacy employee at a pharmacy
- Pharmacy Stock: to model the stock of a certain medication at a pharmacy
- Prescription: to model the prescription written by a GP for a patient
- Prescription Validation: to model a pharmacist validating a certain prescription

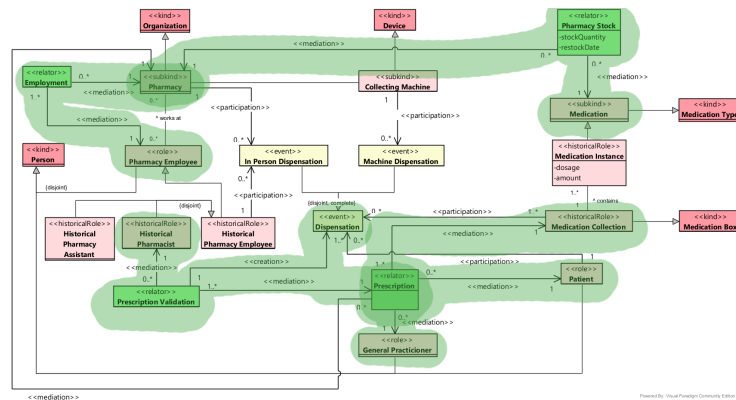


Figure 12: Pharmacy Model - Relator Pattern

Role Pattern (Figure 13) Last, this model contains one event pattern. This is for the event of Dispensation. When a prescription is validated, a Dispensation event happens. This event can either be in person or at a collecting machine, which are both specializing events of the main Dispensation event.

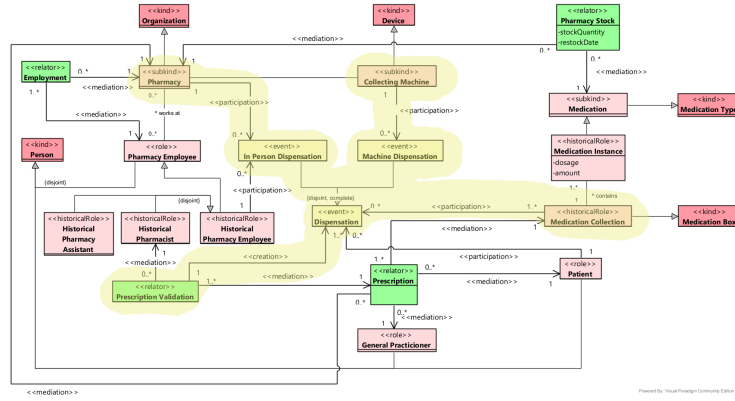


Figure 13: Pharmacy Model - Event Pattern

References

- [1] Thuisarts.nl. How can i ask my GP questions or make an appointment? <https://www.thuisarts.nl/>, Sept. 2021. Accessed: 2025-05-12.
- [2] Zorgwijzer. Healthcare in the Netherlands (Dutch system). <https://www.zorgwijzer.nl/>, Nov. 2024. Accessed: 2025-05-12.